

1 Solution — GIAM 7.3

Problem 7

1.1 Problem

Prove that in a simple graph (an undirected graph with no loops or parallel edges) having n nodes, there must be two nodes having the same degree.

(Implicit: $n \geq 2$, since “two nodes” requires at least two vertices. We return to the $n = 1$ edge case below.)

1.2 Pigeonhole Framing — and Why the Naive Version Fails

Let $G = (V, E)$ be a simple graph with $|V| = n \geq 2$. For each $v \in V$, the *degree* $\deg(v)$ is the number of edges of G incident to v . Since G has no loops and no parallel edges, v is adjacent to at most one of each of the other $n - 1$ vertices, so

$$\deg(v) \in \{0, 1, 2, \dots, n - 1\}.$$

The degree map is then a function

$$d : V \longrightarrow \{0, 1, \dots, n - 1\}.$$

A naive attempt. The domain V has size n ; the codomain $\{0, 1, \dots, n - 1\}$ also has size n . So Theorem 7.3.1 does **not** apply directly — $n > n$ is false. We need a sharper observation to shrink the effective codomain.

1.3 Step 1 — The Endpoints 0 and $n - 1$ Cannot Both Occur

Claim. In any simple graph on $n \geq 2$ vertices, the image $d(V)$ contains *at most one* of the values 0 and $n - 1$.

Proof. Suppose both $0 \in d(V)$ and $n-1 \in d(V)$. Pick $u, v \in V$ with $d(u) = 0$ and $d(v) = n-1$. Note $u \neq v$ since $d(u) \neq d(v)$.

Since $d(v) = n-1$, vertex v is adjacent to *every* other vertex of G — in particular, v is adjacent to u . But then the edge $\{u, v\}$ is incident to u , so $d(u) \geq 1$, contradicting $d(u) = 0$.

Hence $\{0, n-1\} \not\subseteq d(V)$. $\square$

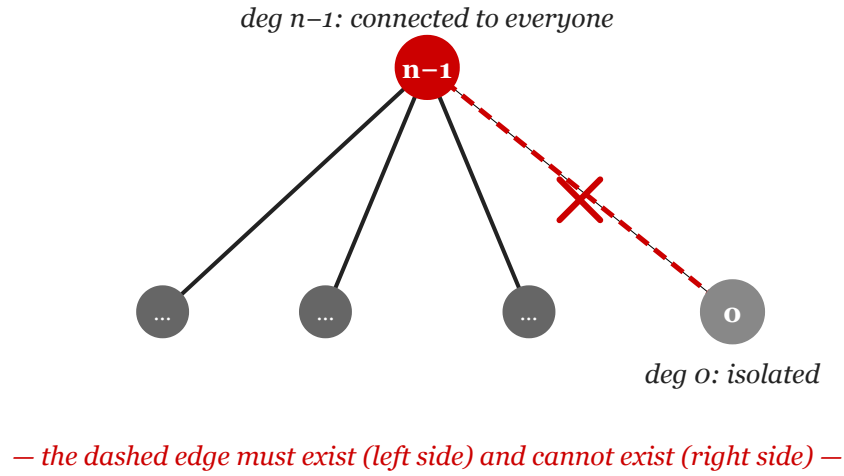


Figure 1.1: A $\deg\text{-(}n\text{-1)}$ vertex is adjacent to every other vertex. If a $\deg\text{-0}$ (isolated) vertex existed, the edge between them would have to exist (forced from the $\deg\text{-(}n\text{-1)}$ side) and cannot exist (forbidden from the $\deg\text{-0}$ side) — a contradiction.

1.4 Step 2 — The Effective Codomain Has Size $\leq n-1$

By Step 1, $d(V)$ misses at least one of $\{0, n-1\}$, so

$$d(V) \subseteq \begin{cases} \{0, 1, \dots, n-2\} & \text{if } n-1 \notin d(V), \\ \{1, 2, \dots, n-1\} & \text{if } 0 \notin d(V). \end{cases}$$

Both sets have size $n-1$. So $|d(V)| \leq n-1$.

1.5 Step 3 — Apply the Pigeonhole Principle

Replace the codomain of d by the image to get the surjection

$$d : V \twoheadrightarrow d(V), \quad |V| = n, \quad |d(V)| \leq n - 1.$$

Since $n > n - 1$, Theorem 7.3.1 tells us d is not injective: there exist distinct $u, v \in V$ with

$$\deg(u) = \deg(v).$$

That is, *two vertices of G share a degree.* ■

1.6 A Worked Example — $n = 4$

The picture below shows a simple graph on $V = \{A, B, C, D\}$ with edges

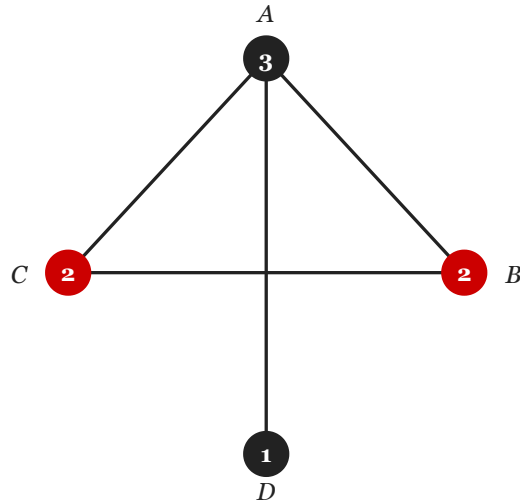
$$E = \{AB, AC, AD, BC\}.$$

Computing degrees:

Vertex	Incident edges	Degree
A	AB, AC, AD	3
B	AB, BC	2
C	AC, BC	2
D	AD	1

Table 1.1

Degree multiset: $\{1, 2, 2, 3\}$. Possible degrees in $\{0, 1, 2, 3\}$ ($= \{0, 1, \dots, n - 1\}$ for $n = 4$) — but $0 \notin d(V)$ since A has degree $n - 1 = 3$, forcing every other vertex to have degree ≥ 1 . The image $d(V) = \{1, 2, 3\}$ has size $3 = n - 1$, and the four vertices map into it — pigeonhole forces a collision. Here B and C both have degree 2.



degree multiset: $\{1, 2, 2, 3\}$ — two vertices share degree 2

Figure 1.2: A 4-vertex simple graph with edge set $\{AB, AC, AD, BC\}$. Vertices are labeled by name (A, B, C, D) with degree numbers inside the circles. Vertices B and C share degree 2 (highlighted in red); A has degree 3 and D has degree 1.

1.7 The Edge Case $n = 1$

For $n = 1$, the graph has a single vertex of degree 0. The statement “two nodes have the same degree” is vacuously *false* (there are not even two nodes). The original claim is implicitly about $n \geq 2$, and the proof above operates in that regime.

(For $n = 2$, the proof goes through: possible degrees in $\{0, 1\}$, but both cannot coexist — a degree-1 vertex needs an edge to *some* other vertex, which would not be isolated. So either both vertices have degree 0 or both have degree 1, and they share. ✓)

1.8 Remark — Why “Simple” is Essential

The proof uses the hypothesis “no loops, no parallel edges” via the upper bound $\deg(v) \leq n - 1$. Dropping either assumption breaks the argument — and the conclusion.

With parallel edges (multigraph). Take $n = 3$ with vertices $\{A, B, C\}$, one edge AB , and two edges BC . Degrees: $A : 1, B : 3, C : 2$. All distinct. ✗

With loops. Take $n = 3$ with $\{A, B, C\}$, a loop on A , and an edge AC . Each loop contributes 2 to its vertex's degree. Degrees: $A : 2 + 1 = 3, B : 0, C : 1$. All distinct. $\times$

So both restrictions are *individually* necessary for the conclusion to hold.

1.9 Remark — Tightness

The bound "*at least two* vertices share a degree" cannot be sharpened to "*at least three* share a degree" in general. There exist simple graphs on $n \geq 2$ vertices where the degree multiset has exactly one repeated value (occurring twice) and all other degrees are distinct.

Example for $n = 4$: the graph above has degree multiset $\{1, 2, 2, 3\}$ — only the value 2 is repeated (twice).

Example for $n = 5$: take the graph with edges $\{AB, AC, AD, AE, BC, CE\}$. Degrees: $A : 4, B : 2, C : 3, D : 1, E : 2$. Multiset $\{1, 2, 2, 3, 4\}$ — only 2 repeated, exactly twice.

In each case the pigeonhole bound is *exactly* met: n vertices into $n - 1$ possible degree values, with exactly one collision.

1.10 Remark — Alternative Framing via Pairing

The argument can be restated more symmetrically. Pair the possible degree values

$$0 \longleftrightarrow n - 1, \quad 1 \longleftrightarrow n - 2, \quad \dots$$

into pairs $\{k, n - 1 - k\}$. Each pair represents a “vertex and its non-neighbors”: if $\deg(v) = k$ then v is *non*-adjacent to $n - 1 - k$ other vertices.

In a simple graph G , the *complement graph* $\bar{G}$ has $\deg_{\bar{G}}(v) = n - 1 - \deg_G(v)$. So the pairing above swaps the degree functions d_G and $d_{\bar{G}}$. The non-coexistence of 0 and $n - 1$ becomes: G cannot have a “universal” vertex and an “isolated” vertex simultaneously — equivalently, $\bar{G}$ cannot have the same.

This complement perspective also gives a strengthening: *both* the graph G and its complement $\bar{G}$ must have two vertices of the same degree, but the *pairs* of equal-degree vertices may differ between G and $\bar{G}$.

1.11 Remark — A Stronger Statement

A counting-mod-2 argument (Problem 2 in Section 7.2) gives:

In any simple graph, the number of odd-degree vertices is even.

This is *not* the statement we proved here (which is about repeated degree *values*), but it shows the same theme: degree sequences in simple graphs are *constrained* in non-obvious ways. Other classical constraints:

- **Erdős–Gallai theorem.** Characterizes which sequences $(d_1, \dots, d_n)$ arise as degree sequences of *some* simple graph.
- **Havel–Hakimi algorithm.** Constructive realization test for proposed degree sequences.

Our pigeonhole observation is the simplest such constraint: *no simple graph realizes the degree sequence* $(0, 1, 2, \dots, n-1)$. Equivalently, the all-distinct degree sequence is impossible.